CASE STUDY: INTELLIGENT SYSTEMS IN PRODUCTION

REVISED TOPIC

Second Check of AOI Results with Supervised Machine Learning

Binary classification of AOI alarms into true defects and false calls (Siemens production data)

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Team 1 · Guided by Dr. Hamidreza Heidari · 23 October 2026 · github.com/bodhi584/aoi-decision-gate

Too many false alarms after automated inspection

98 %

of the boards flagged by the AOI machine are fine (false calls) in real Siemens production data.

Every flagged board goes to a manual inspection station.

Based on the lecture example (p. 18)

“Siemens – Quality Control and Defect Detection”. Key motivation: fewer manual re-inspections of false calls, while true defects are still caught.

A current problem: Siemens and NVIDIA, Erlangen, 21 Oct 2026

“AI second check of AOI results”: real defect or false call? Target: about 40 % fewer false calls.

Our goal

Classify each AOI alarm as true defect or false call, so that fewer false calls go to manual inspection while at most 1 % of true defects are missed.

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ONE GOAL: FEWER FALSE CALLS TO MANUAL INSPECTION, AT MOST 1 % OF TRUE DEFECTS MISSED

Input → System → Output

INPUT

AOI alarm

one flagged board: about 70 AOI measurements per flagged solder joint

component type, AOI error code

SYSTEM (OURS)

Binary classification

supervised machine learning, trained on past manual-inspection results

classification threshold: recall ≥ 99 %

OUTPUT: IS THE AOI ALARM REAL?

False call

→ skip manual inspection

True defect

→ manual inspection, repair

Key factors

False calls

each one costs operator time at the manual station

Missed defects

a missed defective board reaches the customer

Changes over time

the line changes; a model must keep up

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PROJECT SCOPE

What we will and will not do

Aim

This project aims to reduce false calls after automated optical inspection: a binary classifier decides for each AOI alarm whether it is a true defect or a false call, while at most 1 % of true defects are missed.

Includes

- Data preprocessing: Siemens AOI data
- Training three classifiers
- Day-by-day simulation on recorded data
- Evaluation: recall, specificity, confusion matrix

Out of scope (current phase)

- Real hardware deployment
- Integration with physical AOI machines
- Changing the AOI inspection itself (camera, limits)

Conditions and limitations

- One Siemens line, 132 days
- Measurements anonymised (no physical names)
- Labels from operators may contain errors
- Licence: non-commercial (course use)

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What the project will achieve, and how we measure it

1. Reduce false calls: train three classifiers to separate false calls from true defects.
2. Keep quality: set the classification threshold so that recall of true defects is at least 99 %.
3. Test as in operation: day-by-day simulation on the recorded Siemens data, models trained on earlier days only.
4. Compare the three models and identify the most suitable one.

Requirements (positive class = true defect)

≥ 99 %	recall: true defects classified as true defect (≤ 1 % missed)
≥ 20 %	specificity: false calls classified as false call = manual inspections saved (baseline 6.4 %)

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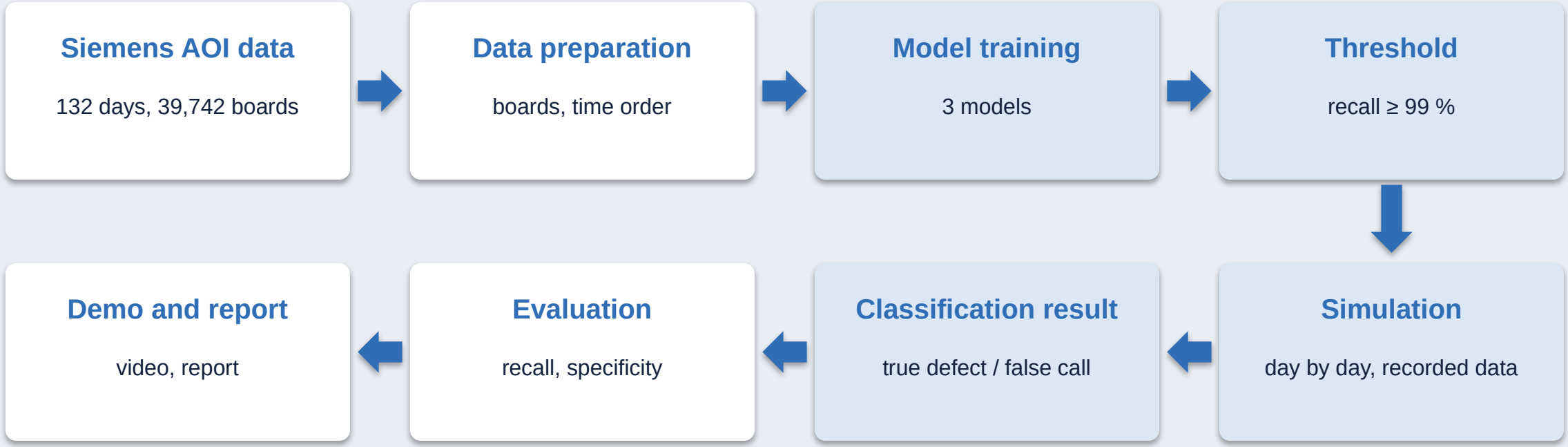
TECHNOLOGY STACK

Data science tools from the lecture (p. 22)

Process	Tool / Framework	Purpose
Data preparation	Python, pandas	Load the Siemens AOI data, group joints into boards, keep time order
Model training	scikit-learn	Logistic regression, random forest, neural network (one hidden layer)
Simulation	Python	Day-by-day simulation on recorded production days (walk-forward)
Evaluation	scikit-learn, pandas	Recall, specificity, confusion matrix
Visualization	matplotlib, Jupyter	Graphical result representation, step-by-step notebook
Code sharing and tests	GitHub, pytest	Repository, tests on every change

PROJECT FLOW CHART

From the Siemens data to the KPIs



Blue boxes: the system we build. White boxes: data and project steps.

WORKING METHODOLOGY

CRISP-DM in four steps

Step 1 · Data preparation

Collected Siemens AOI data: flagged boards, defective & false call

Grouped solder joints into boards; encoded categories

Split data in time order into train, validate, test sets

Step 2 · Model development

Trained logistic regression as baseline

Trained random forest and a neural network (one hidden layer)

Set the classification threshold: recall ≥ 99 %

Step 3 · Simulation integration

Simulated operation day by day on the recorded days

Each day classified by a model trained on earlier days only

Boards classified as false call virtually skip manual inspection

Step 4 · Evaluation and comparison

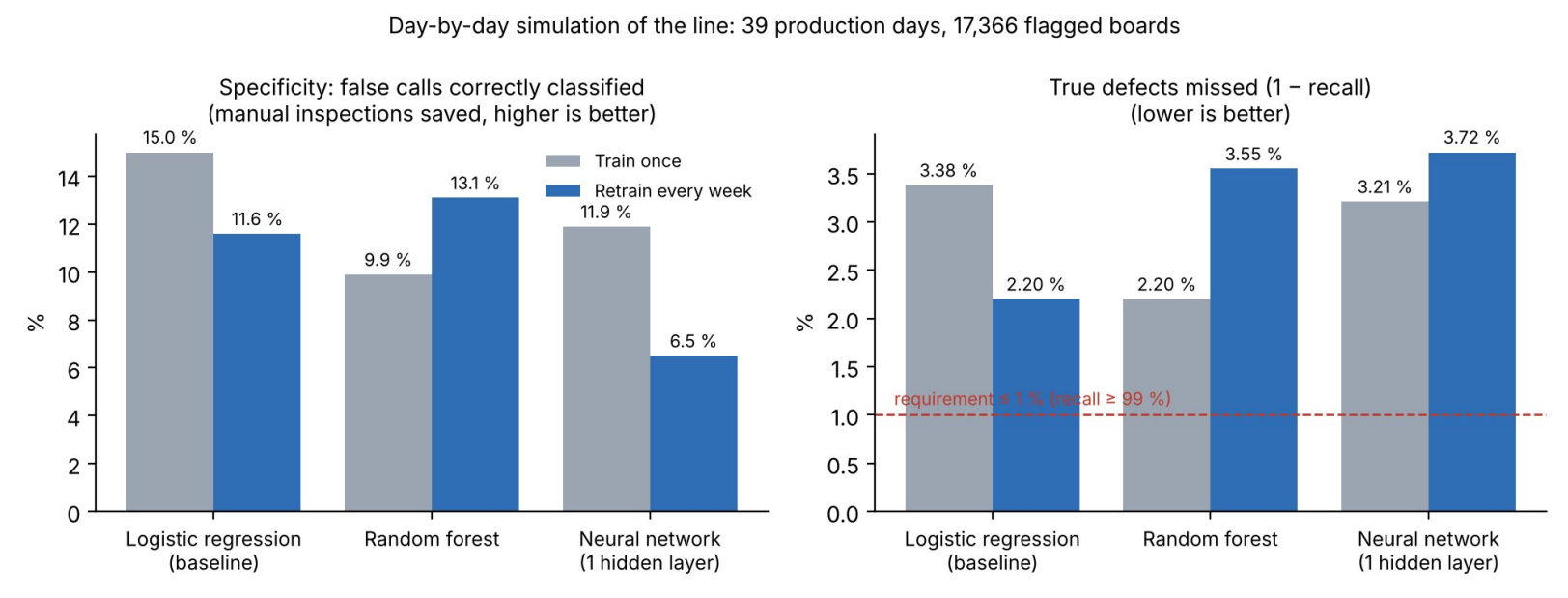
Analyzed recall, specificity and the confusion matrix

Compared the three models to identify the best-performing one

Error analysis; demo video and report

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Day-by-day simulation on recorded data: three classifiers



What it shows

Specificity today: 6.5–15.0 % (false calls correctly classified).

Recall today: 96.3–97.8 % (2.2–3.7 % of true defects missed): below the 99 % requirement.

Next: recall ≥ 99 % with specificity ≥ 20 %.

BILL OF MATERIALS (BOM)

What the project needs

Component	Specification	Quantity	Cost
Data set	Siemens AOI data, Mendeley (CC BY-NC 3.0), 333 MB	1	Free
Computer	Normal laptop or Google Colab (no graphics card needed)	4	Free
Software	Python and open-source libraries	–	Free
Code hosting	GitHub	1 repository	Free

Total cost: 0 €

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EXPECTED OUTCOMES

What we hand in

- ✓ Successfully trained binary classifier for AOI alarms (Python software prototype)
- ✓ Day-by-day simulation on the recorded data, with recall and specificity
- ✓ Comparative performance metrics for the three classifiers
- ✓ Identification of the most suitable model, with limits and error analysis
- ✓ Demo video, run instructions, report, public GitHub repository

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SAFETY & ETHICAL CHECKLIST

Keeping it safe and realistic

- ✔ Simulation-only environment; no physical machinery risk
- ✔ Dataset is anonymized by Siemens (no personal data); used under its licence
- ✔ Ethical principles followed: transparency, fairness, accountability
- ✔ Avoid dataset bias (time-ordered split) and ensure reproducibility of results
- ✔ Safety first in the design: recall of true defects ≥ 99 %

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FUTURE WORK

From the public data set to a real line

1 · Industry partner

An electronics plant shares AOI and manual-inspection records.

2 · AOI interface

Read each board’s AOI result from the machine; send the classification to the manual inspection station.

3 · Shadow mode

The classifier runs next to the operators and only logs its results.

4 · Approval

Quality engineers compare the logs before any board skips manual inspection.

BIBLIOGRAPHY

Sources

Pfab, K., Eichler, R., Mallandur, A., Rothering, M. (2024). Data of automated optical inspection of surface-mounted technology electronic production. Data in Brief 53, 110110.

Siemens (21 Oct 2026). Siemens and NVIDIA advance AI-powered PCB inspection from digital twin to factory. Press release.

Albayari, J., Patra, P., Patel, D., Santos, D. (2025). Enhancing false call detection in automated optical inspection using XGBoost classifier and XAI algorithms. FAIM 2025, Springer.

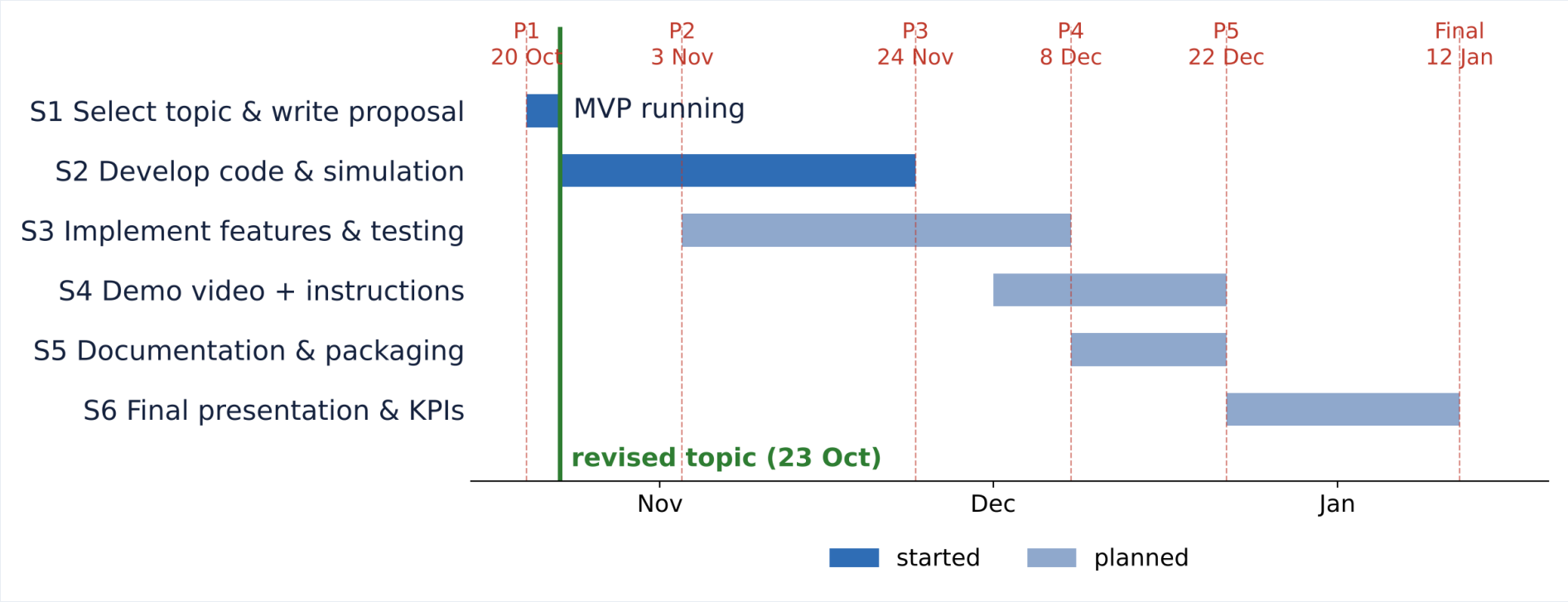
Pedregosa, F. et al. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research 12, 2825–2830.

Chapman, P. et al. (2000). CRISP-DM 1.0: Step-by-step data mining guide. SPSS.

Lecture slides, Case Study Intelligent Systems in Production, p. 8, 12, 18, 22, 33.

MILESTONES

Software & Simulation project, S1–S6 (lecture p. 8, 33)



Repository link

github.com/bodhi584/aoi-decision-gate

Thank you

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